

Integrated Data Center	
Specification	Qty
- The proposed solution should support Role based Access control and Access list to control management access to the device via restrictions based on IP Address, services etc. - The proposed solution must be accessed through the below: - Using the CLI - Using SNMP - REST API - Using the Web Based Management via HTTPs - Virtualization: The proposed solution must have Hypervisor (must not use Open Source) Based Virtualization feature that virtualizes the Device resources—including CPU, memory, network, and acceleration resources. - The Hypervisor used to virtualize the hardware must be a specialized purpose build hypervisor and NOT a commercially available hypervisor (like XEN, VMware, KVM etc.). (Public Available Reference link must be shared) - Each Virtual Instance contains a complete and separated environment of the following: - Resources, - Configurations, - Management, - Operating System - The proposed solution must support 5 Virtual Instance from the day of deployment and support upto 20 Virtual Instances for future Scalability on the same hardware. Using of Open Source/3rd party Network Functions is not recommended. - The proposed solution must support Local Application Switching; Server load Balancing; TCP Multiplexing; HTTP Pooling; HTTP Pipelining; Compression; Caching; TCP Optimization; Filter-based Load Balancing; Transparent Deployments; Content-based Load Balancing; Persistency; HTTP Content Modifications, Web application Firewall, Global server load balancing,SSL offload with TLS Versions and Cipher controls and other latest features. - The proposed solution must support Standalone as well as Virtualized Mode from the day of deployment. - The proposed solution must be capable to handle IPv4 to IPv6 translation and must be IPv6 ready - The proposed solution must have support for HTTP/2 & HTTP/3 Gateway - The proposed solution must support static and dynamic routing protocols like OSPF, RIP1, RIP2, BGP, etc. from the day of deployment. - The proposed solution must support standard VRRP (RFC - 2338) for High Availability purpose. - The proposed solution must have the below metrics: - Minimum Misses, - Hash,	One (01)

○ Persistent Hash, ○ Tunable Hash, ○ Weighted Hash, ○ Least Connections, ○ Least Connections Per Service, ○ Round-Robin, ○ Response Time, ○ Bandwidth, etc • The proposed solution must have Following Functionality: ○ Virtual Matrix Architecture ○ Client Network Address Translation (Proxy IP) ○ Mapping Ports ○ Direct Server Return ○ One Arm Topology Application ○ Direct Access Mode ○ Assigning Multiple IP Addresses ○ Immediate and Delayed Binding • The proposed solution must have framework for customizing application delivery must be provided using user-written scripts, that provides the flexibility to control application flows and fully meet business requirements in a fast and agile manner. • The proposed solution must enable to: ○ Extend existing capacity with delivery of new applications ○ Quickly deploy new services ○ Mitigate application problems without changing the application ○ Preserve existing infrastructure by adding new capabilities without additional equipment • The proposed solution must have DNSSEC based Global Load Balancing from the day of deployment. • The proposed solution must have the Proximity based LLB which monitors 24/7 Full-path transaction completion through Application-aware full-path Health Monitoring module and automatically measures the real-time status of two-way routes between the network and the remote user or servers on the internet based on multiple parameters including latency, packet loss, Cost and load time of the link. • The proposed solution must support: ○ Static NAT ○ Dynamic NAT ○ No-NAT • The proposed solution must support application delivery features of layer 7 load balancing, layer 7 content switch, caching, SSL offload. • The proposed solution must support Client Side RTT, Server Side RTT and overall traffic rendering time on Centralized Dashboard via additional license based external management appliance. • The proposed solution must have the minimum hardware requirement: Traffic Ports support : 2 x 10 GE (supporting 1G Copper/Fibre, 10G Fibre transceivers), 8 x 1 GE RJ45 (without use	
--	--

of Breakout Cable); Device L4 Throughput : 5 Gbps and scalable upto 25 Gbps; Layer 4 connections per second : 500K CPS; Layer 7 requests per second : 700K RPS; Layer 4 Concurrent Connection : 45 M

- The proposed solution must be dedicated purpose built firewall device which enables the convergence of high performing networking and security across the Security Platform. Solution must deliver consistent and context-aware security posture across network with advanced AI/ ML-powered services, application visibility and control, Web security, content security, malware control and DLP.
- The proposed solution must support High-Availability with Active/Active & Active/Passive modes from day-1 on both IPV4 & IPV6.
- The proposed solution should have at least 12x 1GE RJ45 interface, 8 x 1GE SFP slot and 8x 10G SFP+. NGFW (FW + IPS + AVC) throughput must be 10 GBPS for Mix / production traffic or higher from Day 1
- Throughput should be 65 Gbps or higher from the day of deployment
- The proposed solution should have minimum 10 Gbps of Threat Prevention (FW + IPS + AVC + AV) throughput for Mix / production traffic or higher from Day 1
- The proposed solution should have 2000 or more site-to-site & client to site VPN Tunnels.
- The proposed solution should have minimum 2000 concurrent SSL VPN users and should be scalable in future
- The proposed solution should have 500,000 new sessions per second or higher from day 1
- The proposed solution should have at least 8 Million concurrent sessions from day 1
- The proposed solution should support NAT64, DNS64 & DHCPv6
- The proposed solution must support at least 10 Virtual Domains from day 1
- The proposed solution shall be able to operate on either Transparent (bridge) mode to minimize interruption to existing network infrastructure or NAT/Route mode. Both modes can also be available concurrently using Virtual Contexts.
- The proposed solution should have integrated Traffic Shaping functionality.
- The proposed solution should support
 - IPSEC VPN
 - PPTP VPN
 - L2TP VPN
- The proposed solution shall utilize inbuilt hardware VPN acceleration:
 - IPSEC (DES, 3DES, AES) encryption/decryption
 - SSL encryption/decryption
- The proposed solution shall support the following IPSEC VPN

capabilities:

- Multi-zone VPN supports.
 - IPSec, ESP security.
 - Supports NAT traversal
 - Supports Hub and Spoke architecture
 - Supports Redundant gateway architecture
- The proposed solution shall support 2 forms of site-to-site VPN configurations:
 - Route based IPSec tunnel
 - Policy based IPSec tunnel
- The proposed solution shall support IPSEC site-to-site VPN and remote user VPN in transparent mode.
- The proposed solution shall provide IPv6 IPSec feature to support for secure IPv6 traffic in an IPSec VPN.
- The proposed solution appliance must have at least 10 virtual firewall domains /instants active from the day of deployment scalable to 200 (with additional license) with each firewall domains/instances having a separate administrative control OR equivalent, Security zones and VLAN. Each virtual domain must support all the functionalities asked for overall solution.
- The proposed solution should have integrated Antivirus solution
- The proposed system should be able to block, allow or monitor only using AV signatures and file blocking based on per firewall policy based or based on firewall authenticated user groups with configurable selection of the following services:
 - HTTP, HTTPS
 - SMTP, SMTPS
 - POP3, POP3S
 - IMAP, IMAPS
 - FTP, FTPS
- The proposed solution should be able to detect and prevent advanced Malware, Zero-day attack, spear phishing attack, drive by download, watering hole and targeted Advanced Persistent Threat without relying on just Signature database.
- The proposed solution should be able to perform dynamic real-time analysis of advanced malware on the appliance itself to confirm true zero- day and targeted attacks. Cloud infrastructure system for analysis and detection of Malware.
- The proposed solution should be able to block or allow oversize file based on configurable thresholds for each protocol types and per firewall policy.
- The proposed solution should have integrated Web Content Filtering solution without external solution, devices or hardware modules.
- The proposed solution should be able to enable or disable Web Filtering per firewall policy or based on firewall authenticated user groups for both HTTP and HTTPS traffic.
- The proposed solution shall provide web content filtering features:
 - which blocks web plug-ins such as ActiveX, Java Applet, and

Cookies.

- Shall include Web URL block
- Shall include score based web keyword block
- Shall include Web Exempt List
- The proposed solution shall be able to queries a real time database of over 110 million + rated websites categorized into 70+ unique content categories.
- The proposed solution shall have the ability to detect, log and take action against network traffic based on over 4000 application signatures
- The proposed solution should update the application signatures manually or automatically
- The proposed solution should able to define application control list based on selectable application group and/or list and its corresponding actions by the administrator
- The proposed solution should support Active/Standby and Active/Active failover and should not be based on stacking units in clustering.
- The proposed solution should support ether channel or equivalent functionality for the failover control and providing additional level of redundancy.
- The proposed solution should support redundant interfaces to provide interface level redundancy before device failover.
- The proposed solution should support 802.3ad Ether channel or equivalent functionality to increase the bandwidth for a segment.
- The proposed solution should have integrated redundant power supply.
- The proposed solution should have 240 GB of Hard Drive Capacity for logging and reporting.
- The proposed solution should have real-time display of information allows to follow real-time trends in network usage such as the source IP address and the destination URL for HTTP traffic.
- The proposed solution should have ability to configure, manage and monitor NGFW using CLI and GUI both without central management solution. The solution should have full feature parity of centralized management tool on the Next Generation Firewall itself, in case there is any connectivity issues between Next Generation Firewall and central management tool or failure in central management tool
- The proposed solution should have inbuilt feature for two factor authentication for at least two admin users access.
- Two quantities of the above solutions are required
- The proposed solution must comprise of IPS Detection, IPS Mitigation (Detection and Mitigation can be proposed on the same hardware appliance) and External Centralized Management and monitoring system.
- The proposed solution should be a Dedicated appliance (NOT a part/added functionality over Router, UTM, Application Delivery Controller, Proxy based architecture or any StateFul Appliance)

• The proposed solution should support In-Line, SPAN Port, Out-of-Path deployment modes. • The proposed solution should support Protection from SSL Based Attack. • The proposed solution should protect against Zero Day intrusion Attacks and known intrusion attacks. Zero day attack protection should be provided using behavior based technology. • The proposed solution should defend against zero-day network-flood attacks, detect traffic anomalies and prevent zero-day, unknown, flood attacks by identifying the footprint of the anomalous traffic. • The proposed solution must have protection in: ○ TCP floods—which include SYN Flood, TCP Fin + ACK Flood, TCP Reset Flood, TCP SYN + ACK Flood, and TCP Fragmentation Flood ○ UDP flood ○ ICMP flood ○ IGMP flood • The proposed solution must have DNS protection for each type of query including, A, MX, PTR, AAAA, Text, SOA, NAPTR, SRV etc. • The proposed solution must be able to detect and block SYN Flood attacks and should support different mechanism: ○ SYN Protection - Safe Reset ○ SYN Protection - TCP Reset. • The proposed solution should support DNS Challenge and DNS Rate Limit. • The proposed solution should support HTTP Challenge Response without Scripts • The proposed solution should have SIP Flood Protection, UDP and UDP Fragmented Flood. • The proposed solution must identify Server-based vulnerabilities: ○ Web vulnerabilities ○ Mail server vulnerabilities ○ FTP server vulnerabilities ○ SQL server vulnerabilities ○ DNS server vulnerabilities ○ SIP server vulnerabilities ○ Worms and viruses ○ Trojans and backdoors ○ IRC bots ○ Spyware ○ Phishing ○ Anonymizers • The proposed solution should have String Match Engine to support 5000+ inbuilt IPS Signatures apart from custom Signatures. • The proposed solution should have below Security Protection Profiles: ○ Automated IPS Protection. ○ DNS Protections..	
---	--

○ SYN-Flood Protection. ○ Traffic Filters. ○ Anti-Scanning Profile. • The proposed solution should support horizontal and vertical port scanning behavioral protection. Legitimate throughput handling: 5 Gbps from day of deployment and scalable upto 10Gbps. Attack Concurrent Sessions : Unlimited. Inspection Ports supported : 8 X 1 Copper Ports with provision of Bypass, 4 X 10G with provision of Bypass. Latency should be less than 60 microseconds. • The proposed solution should have dedicated Management Port • The proposed solution must capture network traffic in full range (L2-L7 layers) and saves the captured network traffic in a PCAP file. PCAP files can be downloaded from the web user interface and further analyzed in third-party tools. • The proposed solution must be capable of network traffic capture in high-speed networks with speeds of up to 100 Gb/s without any packet loss. Up to 500 Mbps from the whole network traffic can be captured and stored to disk. • The proposed solution must extend the functionalities of probes and is controlled centrally from a collector through a web interface. The web interface is used for entering requests for network traffic capture. It allows the user to define on which probes and their monitoring interfaces the network traffic will be captured. • The proposed solution must extend the functionalities of probes and equips them with a web interface. The web interface is used for entering requests for network traffic capture. It allows the user to define on which probe's monitoring interfaces the network traffic will be recorded. • The proposed solution must allow defining custom filtering criteria for individual capture requests to capture specific parts of network traffic. Filtering criteria are parameters from layer L2-L4 and L7. • The proposed solution must allow filtering network traffic by MAC addresses, VLAN tags and MPLS labels. • The proposed solution must allow filtering network traffic by IPv4, IPv6 addresses of hosts or whole networks. • The proposed solution must allow filtering network traffic by TCP, UDP and SCTP ports. • The proposed solution must allow filtering network traffic of VoIP calls using the SIP or H.323 protocol. • The proposed solution must allow defining the time interval of traffic recording for individual requests. Capture tasks can be scheduled. • The proposed solution must allow defining groups of users who will have access rights to download the traffic record when entering a new capture request. • The proposed solution must allow network traffic recording which can be automatically initiated upon event detection by the system for automatic NetFlow analysis. • The proposed solution must allow defining on which probes and their monitoring interfaces the network traffic will be recorded.	
--	--

- The proposed solution must allow the analysis of network traffic and presents analysis results in an understandable way in the form of events. Events describe individual communications from start to the end. Events are separated to different levels to describe how severe issues were detected and indicated issues in the network traffic for the set of supported protocols. System allows to analyze own traffic recordings uploaded to the system in the form of a PCAP file.
- The proposed solution must support analysis that can be started automatically after the traffic recording or after the external PCAP file is uploaded to the system.
- The proposed solution must have Network traffic analysis detail which presents users with comprehensible visualization of individual analyzed communications in form of an event tree. Event trees correspond to individual communications and events describe particular communication from the start to the end. Each event provides detail which contains information about communication peers, used ports, time and other information from packet data including protocol data. In the case of non-standard states or other errors, the user is presented with a detailed description of the issues including translation of error codes.
- The proposed solution must contain built-in expert knowledge which is used to suggest users with remedial actions for all detected non-standard or error events.
- The proposed solution must allow to filter analysis results to quickly identify problematic communications.
- The proposed solution must provide a documented API for retrieving records of captured network traffic. The API also allows entering traffic capture requests and defining the capture time period and filter criteria.
- The proposed solution must have automatic rotation of old data to free disk space for new recordings.
- The proposed solution must provide a time and capacity limited buffer to keep short term full packet history in case that these packets will be subject to packet capture (typically together with automated traffic recording capability).
- The proposed solution should have minimum 4x 1/10G SFP interfaces populated with 2*10G SR SFP, 2*10G LR SFP & 4* RJ45 ports
- The proposed solution should have additional interface modules/slot support for future expansion of onboard interfaces
- The proposed solution should support port ACL with I2, L3 and L4 parameters
- The proposed solution should support LLDP and LACP to bundle links and detect miscabling issues.
- The proposed solution should support Routing Protocols: OSPFv2, BGP, IS-IS, and RIPv2
- The proposed solution should support BFD.
- The proposed solution should support VRRP V4.
- The proposed solution should support VXLAN+EVPN overlay

technology

- The proposed solution should support NAT
- The proposed solution should be able to support 1M or higher IPv4 routes
- The proposed solution should be able to support minimum 128 IPSEC tunnels with AES256 encryption
- The proposed solution should support minimum 5 Gbps encrypted throughput
- The proposed solution should be able to support GRE tunnels
- The proposed solution should be able to work as traditional router and capable of upgrading to SDWAN router with additional license, if required in future
- The proposed solution should support ISSU for software upgrades and hitless patching
- The proposed solution should support maintenance mode/ Graceful insertion and removal (GIR) to isolate device from the network in order to perform debugging or an upgrade while gracefully steering traffic to peer nodes
- The proposed solution should 1+1 redundant & hot-swappable Fans
- The proposed solution should support 1+1 redundant & hot-swappable power with support for both AC and DC power supply options.
- The proposed solution Should support Control Plane protection (CoPP)
- The proposed solution should support port ACL with I2, L3 and L4 parameters
- The proposed solution should support port based DOS protection (PDP)
- The proposed solution should have programmability and automation support with on board bash, python, C++ , GO & OpenConfig.
- The proposed solution Should support in-band telemetry
- The proposed solution should support 8 queues per port
- The proposed solution should support priority queue
- The proposed solution should support ACL based classification for QoS
- The proposed solution should support rate limiting function like policing and shaping
- The proposed solution should have ten network switches & four routers should run on Same OS and should be able to manage from Same Dashboard for simplified operations.
- One unit 42U rack with following specifications
- Height: 73.5 inches (1867 mm)
- Width: Standard widths are 19 inches (482.6 mm), which is the industry standard for networking equipment.
- Mounting Rails:
- Typically, 42U racks have adjustable and square-hole mounting rails that allow flexibility in mounting various types of equipment. Square holes accommodate cage nuts and screws for secure mounting.

- Construction:
- Racks are commonly made of steel or aluminum for durability and strength.
- Doors and side panels may be perforated to facilitate airflow and cooling.
- Weight Capacity:
- Racks are designed to support a certain amount of weight. The weight capacity can vary, but it's usually in the range of several hundred to over a thousand pounds.
- Ventilation:
- Racks may have built-in fans or spaces for fans to ensure proper ventilation and cooling for the equipment inside.
- Doors and Panels:
- Front and rear doors provide physical security and may have locks.
- Removable side panels for easy access to equipment.
- Cable Management:
- Built-in cable management features, such as cable channels and cable tie points, help organize and route cables neatly.
- Grounding:
- Grounding options are provided to ensure proper grounding for equipment.
- Mobility:
- Some racks come with casters for easy mobility, while others are designed for stationary use.
- Rack Units (U):
- The rack should have clearly marked rack units on the front and rear, allowing for easy equipment alignment and installation.
- Compatibility:
- Compliant with industry standards, such as the 19-inch rack standard.
- Color:
- Racks are often black or a neutral color.
- Accessory Compatibility:
- Compatible with various rack accessories such as shelves, cable organizers, and power distribution units (PDUs).
- Grounding and Bonding:
- Proper grounding and bonding provisions for electrical safety.
- Ten quantities of the below mentioned specifications are required
- The proposed solution must support ten instances of multiple platforms of server operating systems i.e. Windows, Linux, RedHat, CentOS, Oracle, Debian, SUSE, Ubuntu, Solaris, AIX, Amazon Linux, Microsoft Windows Server (2012 & 2012 R2, 2016,2019,2022 etc.), Red Hat Enterprise Linux (6,7,8,9 etc.), Solaris (10.0,11.0,11.1,11.2,11.3,11.4), Oracle Linux (6,7,8,9), AIX (7.1,7.2,7.3etc.), CentOS (6,7,8 etc.) and Suse Linux (12,15 etc.)
- The proposed solution should have the following protection mechanism:
 - Anti-Malware

○ Firewall ○ Host Intrusion Prevention System ○ Application Control ○ Integrity Monitoring ○ Sandbox Integration • The proposed solution should have the necessary hardware and software required for the deployment of solution. • The proposed solution must protect against all kinds of viruses, Trojans and worms including but not limited to: boot sector, master boot sector, memory resident, macro, stealth and polymorphism etc.; and any other forms of exploits • The proposed solution shall also protect against certain non-virus threats, such as Spyware, adware, dialers, joke programs, remote Access and hacking tools, which can be used with malicious intent • In the proposed solution, the clients shall be loaded on machines and the server shall distribute the updates to the clients. • The proposed solution should have capabilities to inspect applications, as well as the applications' sub-components (DLLs) as they are executed • The proposed solution should have the ability to detect applications that attempt to spread themselves over the network • The proposed solution must provide application of customized enforcement actions based on malicious file types such as Delete, Block, Quarantine. • The proposed solution should have Post Infection Cleanup capability on detection of a malware infection, the solution should allow removal of traces of malware from the system by cleaning up the following automatically or via remote remediation from a centralized management console : ○ Detected malicious file ○ Affected registry entries ○ Any new files dropped by malware ○ Windows services created by malware ○ Any other system settings affected by malware • The proposed solution should protect documents against unauthorized encryption or modification. • The proposed solution should protect against fileless malware • The proposed solution should provide detection and blocking of Command and control (C&C) traffic • The proposed solution should protect from malware in memory and boot sector on system start-up • The proposed solution should provide real time protection against malware • The proposed solution should be able to add files, folders or extensions to an exclude list from detection • The proposed solution should protect from malware in compressed file formats • The proposed solution should detect access of malicious and dangerous websites	
--	--

- The proposed solution should utilize machine learning on Windows systems to detect malware, which do not have any signatures
- The proposed solution should utilize behavioral techniques on Windows systems to detect malware based on the behavior of the file
- The proposed solution should provide submission of suspicious files to a sandbox / threat- analysis solution for virtual execution / Root-Cause analysis. The sandbox / threat - analysis solution shall provide execution / RCA result for the suspicious file
- The proposed solution should provide on-premise sandboxing capabilities
- The proposed solution should have ability to make firewall rules to filter traffic based on source and destination IP address, port, MAC address, etc.
- The proposed solution should detect reconnaissance activities such as port scans and Solution should be capable of blocking and detecting Ipv4/IPv6 attacks
- The proposed solution should detect vulnerabilities and provide complete visibility of vulnerabilities in operating system with CVE details
- The proposed solution Should provide automatic recommendations and enforcement of HIPS rules against existing vulnerabilities, Solution also allows the automatic removal of HIPS rules if a vulnerability no longer exists - For Example - If a patch is deployed unwanted signatures should be un-assigned
- The proposed solution should also show the available network IPS rules which can be enforced from the network IPS for the identified vulnerabilities on the servers.
- The proposed solution should support CVE cross referencing when applicable for vulnerabilities.
- The proposed solution should detect suspicious network traffic. It shall allow blocking of all traffic from the originating servers
- The proposed solution should have the capability of restrict usage of unauthorized applications and enable blacklist/whitelist of applications on the server
- The proposed solution should provide real time integrity monitoring of critical operating system and application elements such as directories, files, registry keys and values to detect and report suspicious activity such as modifications
- The proposed solution provide auto-recommendation scans to enforce the integrity monitoring rules which are applicable for the servers
- The proposed solution should have an updating mechanism using local update server
- The proposed solution should provide a single unified management dashboard to view the status of installed agents
- The proposed solution should offer enterprise-wide visibility over the status of all the deployed components from a central dashboard. The dashboard shall provide a summarized view to analyze top

threats & summary of malware traffic or any other threats • The proposed solution should offer creation of granular policies based on flexible attributes which can be deployed with consistent policy enforcement across distributed environments and multiple components from one management platform • The proposed solution should offer a central repository of the updates that can be distributed to the managed components • The proposed solution should be able to receive logs from the managed servers and store them centrally • The proposed solution should provide alerts to users in case of any security incident • The proposed solution must provide notifications for important events. The notifications must be send through email or SNMP traps • The proposed solution should support multiple administrator/full access roles accounts. Each administrator account shall be configurable with the desired level of management privileges for different components. • The proposed solution should provide logging of administrative activities performed by the administrators • The proposed solution should be capable of providing detailed reports containing data from all the deployed components • The proposed solution should have ability to Export reports to PDF or text format as per availability. • The proposed solution should be able to work with connected threat defense system wherein solution should be able to share the suspicious objects/detected IOCs for the unknown threats which are analyzed in sandboxed environment with other integrated solution like endpoint security solution, NGIPS and APT. • The proposed solution should be supplied with all necessary licenses and subscriptions including Antimalware, Host IPS, Application Control, File Integrity monitoring, Central management etc. • The proposed solution should include all type of cables, adaptors etc. as required to implement for the same. • The proposed solution should cover three years comprehensive warranty.	
---	--